

Word Search Using Shared Memor

Problem Description

Implement a C program that searches for words within a matrix of letters read from a text file. The program must create **four child processes**, each responsible for searching in a specific direction:

- **Child 1:** searches **to the right**
- **Child 2:** searches **to the left**
- **Child 3:** searches **downward**
- **Child 4:** searches **upward**

The parent process coordinates the children and stores the search results in a **shared memory segment**. Once all children finish, the parent process will print a consolidated report showing where each word was found.

Input File Format

The input file (e.g., `matrix.txt`) will have the following structure:

```
n
m
#matrix values
k
#words
```

Example:

```
4
5
A D C T E
F O H A J
K G M C O
P Q R S T
3
DOG
CAT
BAT
```

Where:

- `n` = number of rows
- `m` = number of columns

- Followed by `n` rows of `m` characters each
 - Then `k` = number of words to search for
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Functional Requirements

1. The **parent process** must:
 2. Read the input file and load the matrix and word list into memory.
 3. Create **shared memory** to store the matrix and the search results.
 4. Create **four child processes** using `fork()`.
 5. Wait for all child processes to finish.
 6. Print a final report showing all found words with their positions and directions.
 7. Each **child process** must:
 8. Access the matrix and words through the shared memory.
 9. Search for all words **in only one assigned direction**.
 10. Write its results in the shared memory structure, specifying:
 - Word found
 - Starting position (row, column)
 - Direction (up, down, left, right)
 11. The program must ensure:
 12. Children **do not overwrite** each other's results (using a flag or basic synchronization logic).
 13. No communication through pipes or temporary files — **only shared memory**.
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Expected Output Example

```
Word: DOG found at (0,2) going down
Word: CAT found at (2,3) going up
Word: BAT not found
```